Math 220: Homework 4

Lance Remigio

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Problem 1 (Closure and Boundary of Balls in a Metric Space)**.** Let (X, d) be a metric space, $x \in X$, $r > 0$.

(i) Prove that $\overline{B_r(x)} \subseteq \{y \in X : d(y, x) \leq r\}$.

Proof. Let $z \in \overline{B_r(x)}$ be arbitrary. Then by definition of $\overline{B_r(x)}$, it follows that either $z \in B_r(x)$ or $z \in D(B_r(x))$. Suppose $z \in B_r(x)$. Since $B_r(x) \subseteq \{y \in X : d(y, x) \leq r\}$, it follows that $z \in \{y \in X : d(y, x) \leq r\}$ and we are done. Now, suppose $z \in D(B_r(x))$. Then it follows by definition that z is a limit point of $B_r(x)$; that is, for all open sets U, $U \cap (B_r(x) \setminus \{z\}) \neq \emptyset$. Suppose for contradiction that $z \notin \{y \in X : d(y, x) \leq r\}$. Thus, $d(z, x) > r$ and so clearly $z \notin B_r(x)$. Also, since $z \in B_r(x)$, it also follows that $B_r(x) \setminus \{z\} = B_r(x)$. Hence, we see that

$$\text{for all open sets } U,\ U \cap B_r(x) \neq \emptyset.$$

However, since X is a metric space, we know that it must satisfy the Hausdorff property. Since x and z are two distinct points in X, there must exists another open set V such that $z \in V$ (some open ball with center z in this case) where $V \cap B_r(x) = \emptyset$. But this contradicts our assumption that $U \cap B_r(x) \neq \emptyset$ for all open sets U in X. Thus, $z \in \{y \in X : d(y, x) \leq r\}$ and we are done. ∎

(ii) Is it true that $\overline{B_r(x)} = \{y \in X : d(y, x) \leq r\}$. Give a proof or counterexample, as appropriate.

Solution. We will show a counterexample. Let X be the discrete metric space. By definition, we see that

$$B_r(x) = \{y \in X : d(y, x) < r\}.$$

Inside the discrete metric space, $B_r(x)$ is also a closed set. Hence, $B_r(x) = \overline{B_r(x)}$. However,

$$\overline{B_r(x)} = B_r(x) \neq \{y \in X : d(y, x) \leq r\}.$$

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(iii) Prove that $\partial B_r(x) \subseteq \{y \in X : d(y, x) = r\}$.

Proof. Let $z \in \partial B_r(x)$. Since $\partial B_r(x) = \overline{B_r(x)} \cap \overline{[B_r(x)]^c}$, it follows that $z \in \overline{B_r(x)}$ and $z \in \overline{[B_r(x)]^c}$. Since $\overline{[B_r(x)]^c} = \left[\text{int}(B_r(x))\right]^c$, it follows that

$$z \in \overline{B_r(x)} \text{ and } z \in \left[\text{int}(B_r(x))\right]^c.$$

Also, $\overline{B_r(x)} = B_r(x) \cup D(B_r(x))$. Hence, either $z \in B_r(x)$ or $z \in D(B_r(x))$, and $z \in \left[\text{int}(B_r(x))\right]^c$. However, since $z \in \left[\text{int}(B_r(x))\right]^c$, it follows that $z \notin \text{int}(B_r(x))$. That is,

$$\text{For all open sets in } U \text{ containing } z,\ U \cap B_r(x) = \emptyset.$$

Hence, $U \subseteq [B_r(x)]^c$. Hence, $z \in [B_r(x)]^c$ and so by definition, $d(x, z) \geq r$. This forces $z \in D(B_r(x))$, otherwise if we assume that $z \in B_r(x)$, then we will get a contradiction.

Hence, z is a limit point of $B_r(x)$. That is, for all open sets U in X containing z, we have

$$U \cap (B_r(x) \setminus \{z\}) \neq \emptyset.$$

Since X is a metric space, the above translates to (in terms of open balls) as

$$\text{For all } \varepsilon > 0,\ B_\varepsilon(z) \cap (B_r(x) \setminus \{z\}) \neq \emptyset.$$

Since $z \notin B_r(x)$, we get that $B_r(x) \setminus \{z\} = B_r(x)$. Hence, the above can be re-written as

$$\text{For all } \varepsilon > 0,\ B_\varepsilon(z) \cap B_r(x) \neq \emptyset.$$

In particular, for $\varepsilon = r > 0$, it follows that $B_r(z) \cap B_r(x) \neq \emptyset$. The only option here would be to have $d(x,z) = r$ and so we are done.

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(iv) Is it true that $\partial B_r(x) = \{y \in X : d(y,x) = r\}$. Give a proof or counterexample, as appropriate.

Solution. Let X be the discrete metric space. Note that the only element that is not in $\{y \in X : d(y,x) = r\}$ is x itself because if it was, then $d(y,x) = 0$ would imply that $r = 0$ contradicting the $r > 0$. Hence, $\{y \in X : d(y,x) = r\} = X \setminus \{x\}$. But $\partial B_r(x) = \emptyset$ and so clearly

$$\partial B_r(x) \neq X \setminus \{x\} = \{y \in X : d(y,x) = r\}.$$

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